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Data Warehousing

Course Instructor

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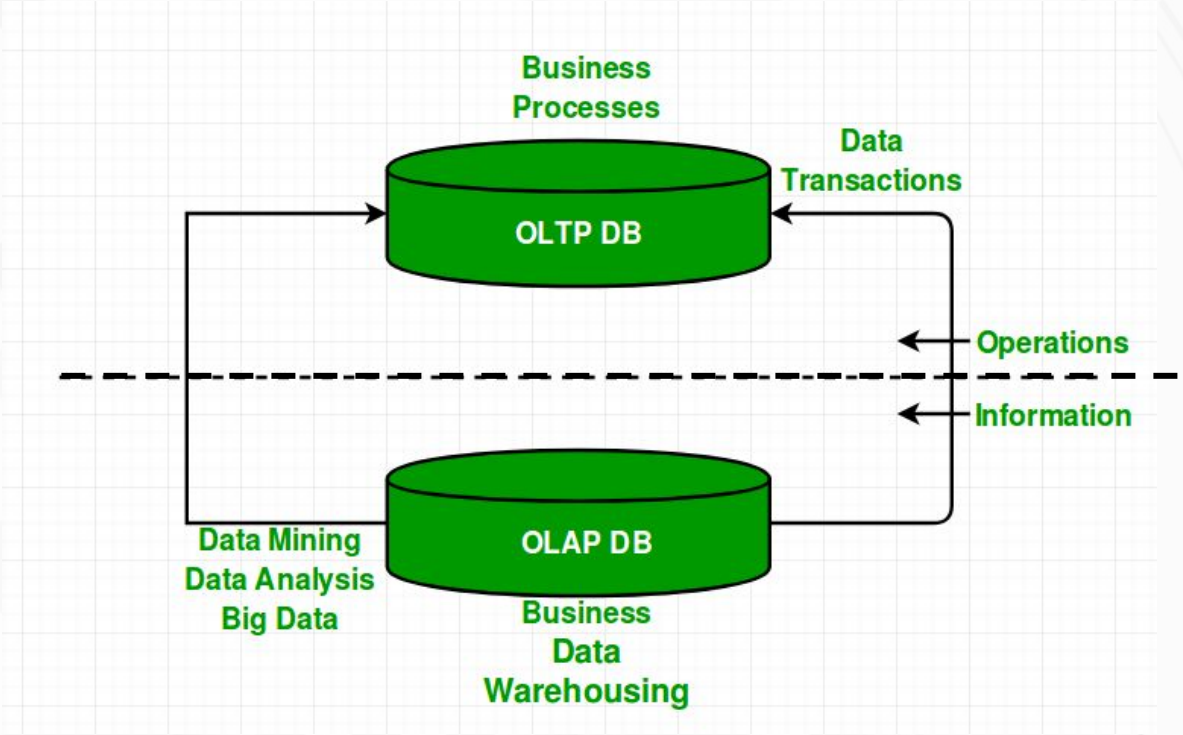
DATA WAREHOUSING : **OLAP & OLTP**



A database that is used to support day-to-day business transactions is called an OLTP (Online Transactional Processing) database.

OLTP relational databases are designed around the need to quickly perform four actions: *Create, Retrieve, Update* and *Delete*. The anagram CRUD is often used for these four requirements.

A database that is used to support day-to-day business reporting is called an OLAP (Online Analytical Processing) database.

Online analytical processing (OLAP) is software technology you can use to analyze business data from different points of view. Organizations collect and store data from multiple data sources, such as websites, applications, smart meters, and internal systems.

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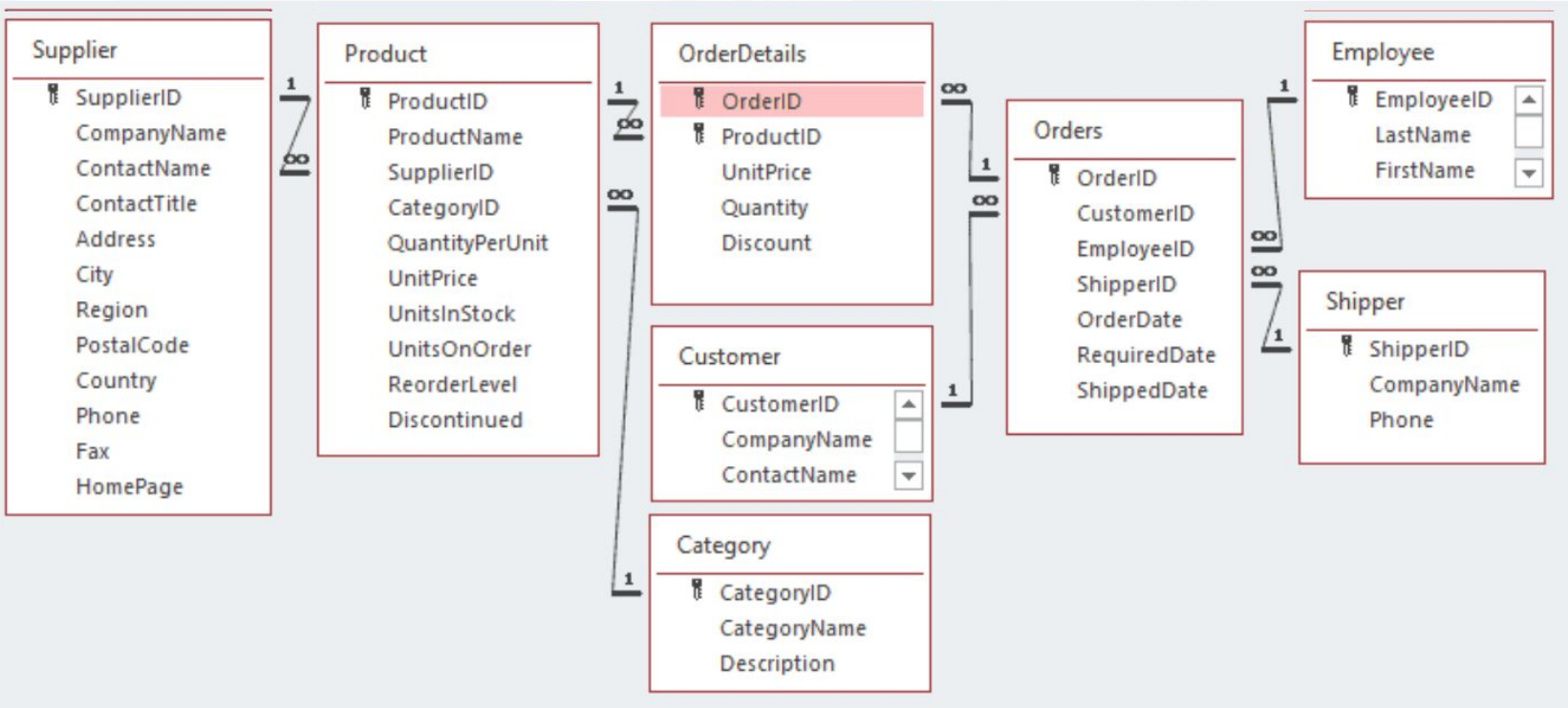
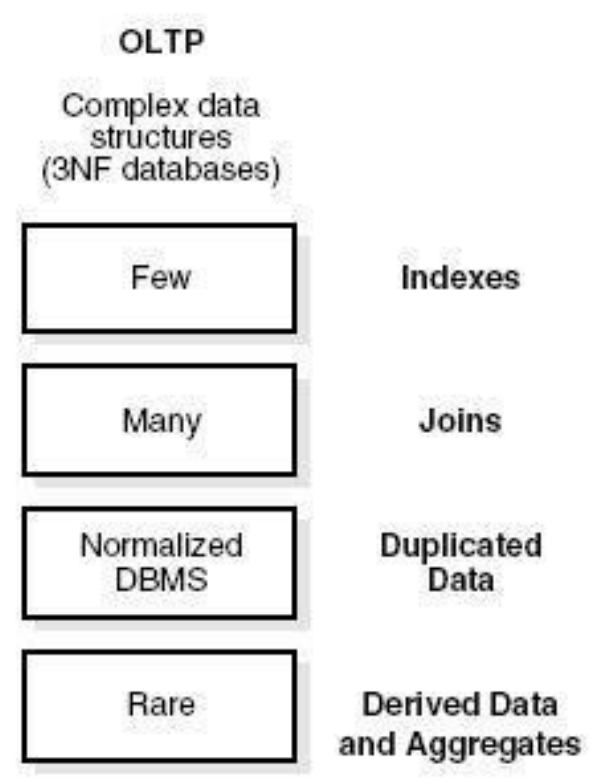
DATA WAREHOUSING : **UNDERSTANDING OLTP FIRST**

OLTP systems are the beginning point for data that will end up in a warehouse. In this sense, OLTP systems could be client/server databases, legacy mainframe systems, and stand-alone file-based PC applications. The focus of these systems is the details of business transactions.

Imagine that your computer application manages employee records (stored in a database). If an employee's details change you would need to first *Retrieve* (find) the employee's record. You'd then have to *Update* the record with the new information.

If an employee left the company, you might want to *Delete* their record. If a new employee was then recruited, you'd need to *Create* a new employee record.





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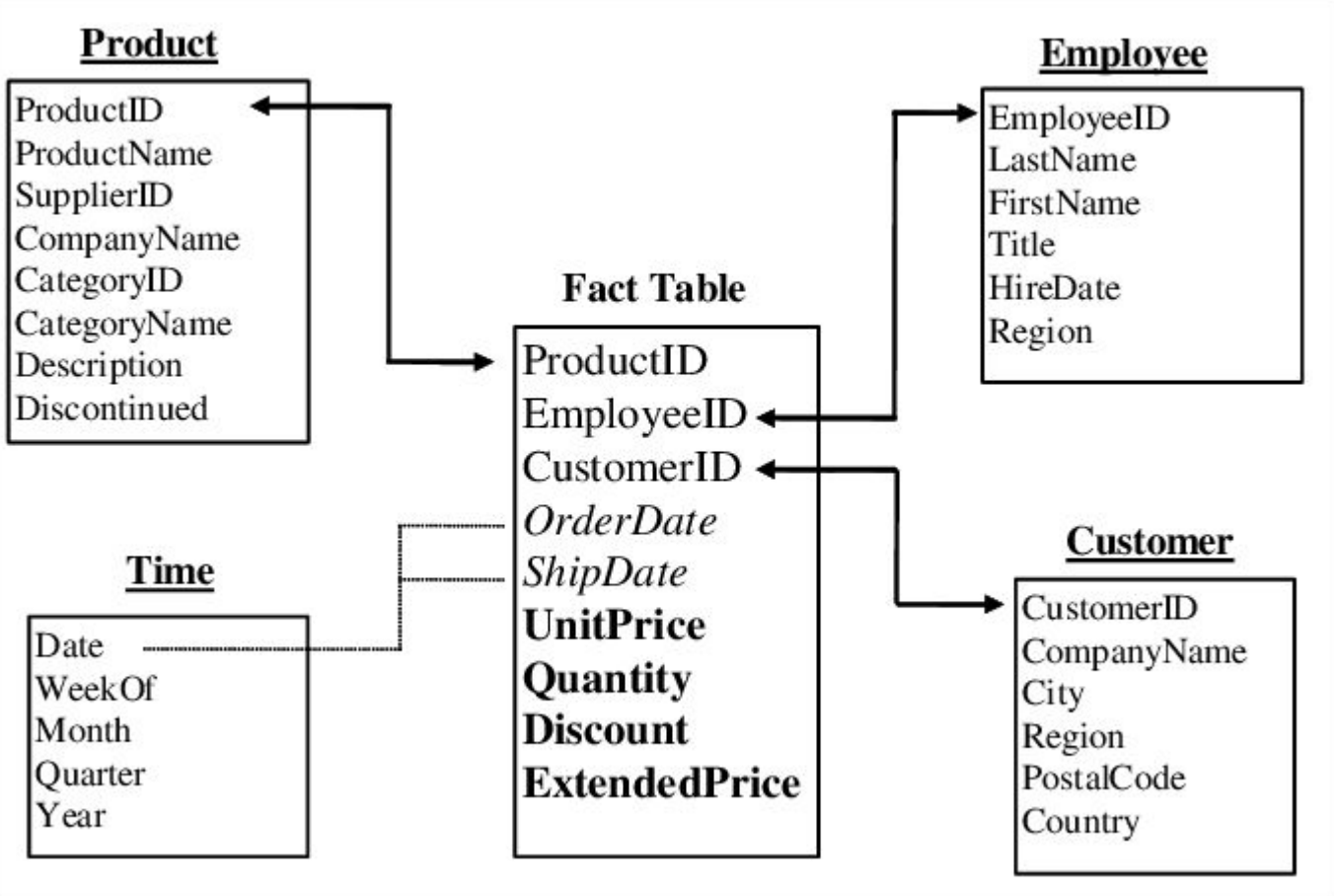
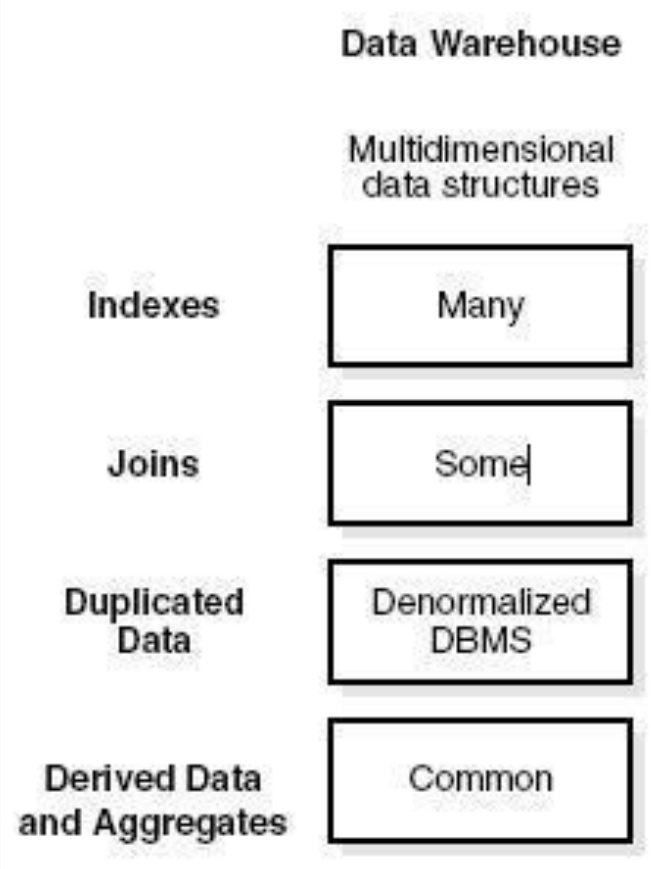
DATA WAREHOUSING : **UNDERSTANDING OLAP SECOND**

OLAP data is moved into a denormalized database, which is your data warehouse. The warehouse may then be split into data marts. You should notice the differences between browsing data stored in a transaction processing system and browsing that in a data warehouse.





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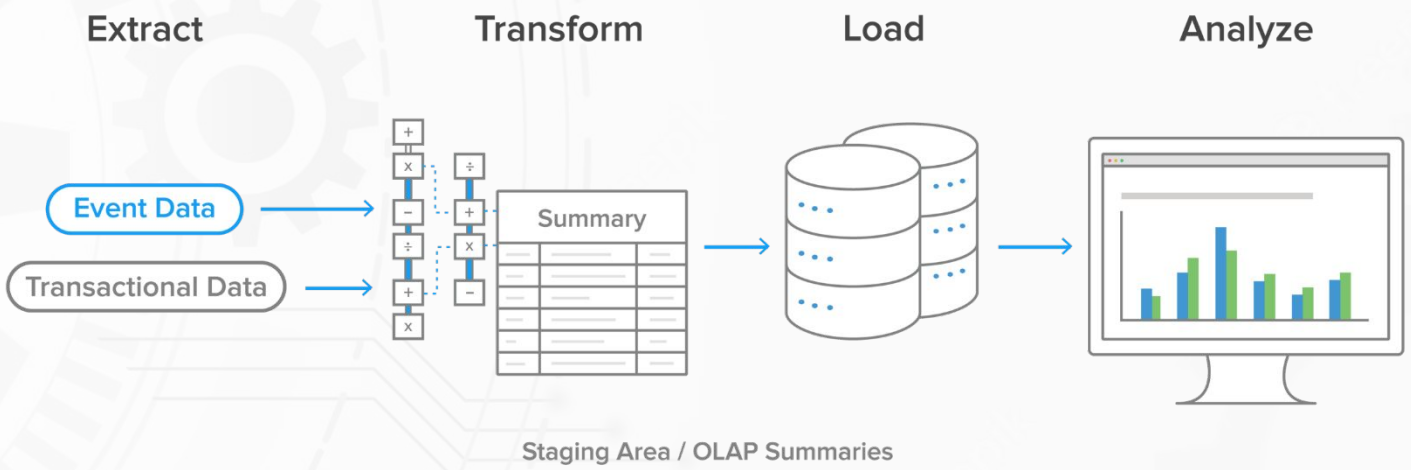
DATA PROCESSING : **EXTRACT, TRANSFORM & LOAD**

ETL is a type of data integration process referring to three distinct but interrelated steps (Extract, Transform and Load) and is used to synthesize data from multiple sources many times to build a Data Warehouse, Data Hub, or Data Lake. A basic ETL process can be categorized in the below stages:

1. Data Extraction
2. Data Cleansing
3. Transformation
4. Load



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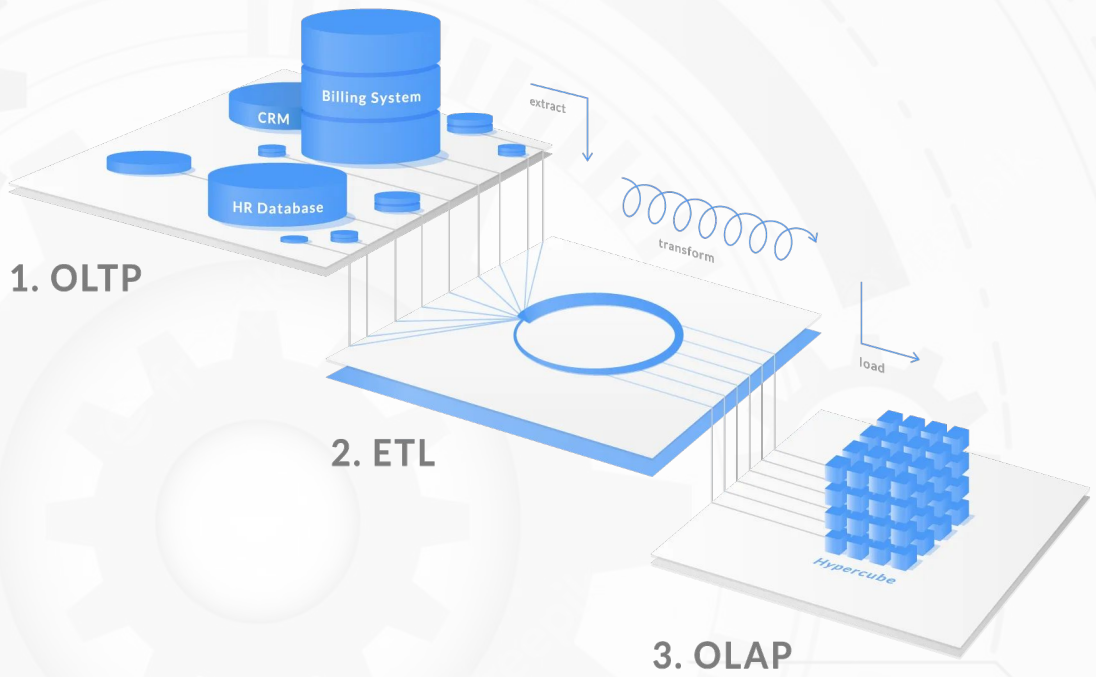
DATA WAREHOUSING: **PUTTING IT ALL TOGETHER**



We now know that:

OLTP is used to store and manage data for day-to-day operations;

OLAP is used to analyze that data.

This is exactly how they are used in an existing business.

Now OLAP is for Reporting and we will see Characteristics of reporting such as viewing current as well as historical data and analyzing trend and change analysis.

Hence we will be having transactions AKA Business events recorded in Facts in OLAP and Business Stakeholders such as Customers, Suppliers, Products and Stores as Dimensions.

This is where Dimensional Modelling Begins.

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DIMENSIONAL MODELLING: **INTRODUCTION**

Before we delve into the details of dimensional modelling, it is helpful to focus on the fundamental goals of data warehousing and business intelligence. The goals can be readily developed by walking through the halls of any organization and listening to business management. These recurring themes have existed for more than three decades:

- “We collect tons of data, but we can’t access it.”
- “We need to slice and dice the data every which way.”
- “Business people need to get at the data easily.”
- “Just show me what is important.”
- “We spend entire meetings arguing about who has the right numbers rather than making decisions.”
- “We want people to use information to support more fact-based decision making.”

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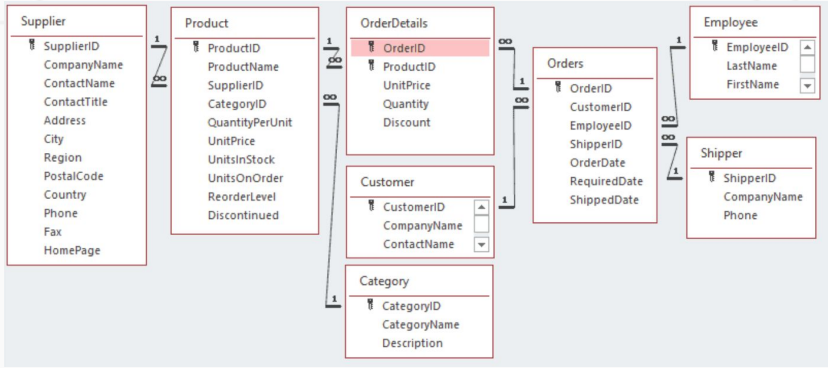
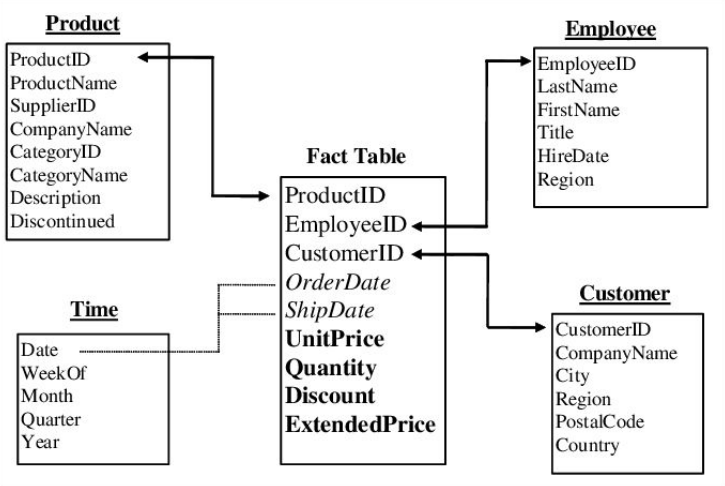
DIMENSIONAL MODELLING: **INTRODUCTION**

The industry sometimes refers to 3NF models as entity-relationship (ER)models. Entity-relationship diagrams (ER diagrams or ERDs) are drawings that communicate the relationships between tables. Both 3NF and dimensional models can be represented in ERDs because both consist of joined relational tables; the key difference between 3NF and dimensional models is the degree of normalization.



Because both model types can be presented as ERDs, we refrain from referring to 3NF models as ER models; instead, we call them normalized models to minimize confusion. Normalized 3NF structures are immensely useful in operational processing because an update or insert transaction touches the database in only one place.

VS



Normalized models, however, are too complicated for BI queries. Users can't under- stand, navigate, or remember normalized models that resemble a map of the Los- Angeles freeway system. Likewise, most relational database management systems can't efficiently query a normalized model; the complexity of users' unpredictable queries overwhelms the database optimizers, resulting in disastrous query performance. The use of normalized modeling in the DW/BI presentation area defeats the intuitive and high-performance retrieval of data. Fortunately, dimensional modeling addresses the problem of overly complex schemas in the presentation area.

NOTE A dimensional model contains the same information as a normalized model, but packages the data in a format that delivers user understandability, query performance, and resilience to change.

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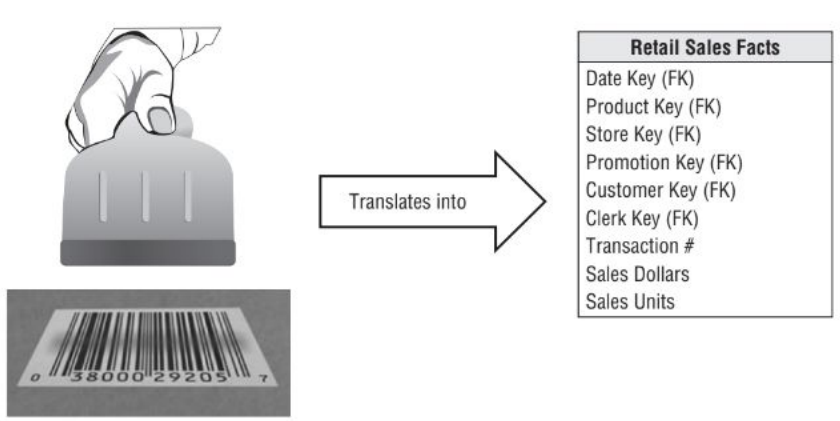
DIMENSIONAL MODELLING: **FACT TABLES**

The **Fact Table** in a dimensional model stores the performance measurements resulting from an organization's business process events. You should strive to store the low-level measurement data resulting from a business process in a single dimensional model. Because measurement data is overwhelmingly the largest set of data, it should not be replicated in multiple places for multiple organizational functions around the enterprise. Allowing business users from multiple organizations to access a single centralized repository for each set of measurement data ensures the use of consistent data throughout the enterprise.

The term fact represents a business measure. Imagine standing in the marketplace watching products being sold and writing down the unit quantity and dollar sales amount for each product in each sales transaction. These measurements are captured as products are scanned at the register, Each row in a fact table corresponds to a measurement event.



The data on each row is at a specific level of detail, referred to as the grain, such as one row per product sold on a sales transaction. One of the core tenets of dimensional modeling is that all the measurement rows in a fact table must be at the same grain. Having the discipline to create fact tables with a single level of detail ensures that measurements aren't inappropriately double-counted.

NOTE: The idea that a measurement event in the physical world has a one-to-one relationship to a single row in the corresponding fact table is a bedrock principle for dimensional modeling. Everything else builds from this foundation. Means if you model like you have Order Fact & Order Detail Fact then Its not correct Dimensional Modelling.

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DIMENSIONAL MODELLING: **FACT TABLES**

If there is no sales activity for a given product, you don't put any rows in the table. It is important that you do not try to fill the fact table with zeros representing no activity because these zeros would overwhelm most fact tables. By including only true activity, fact tables tend to be quite sparse. Despite their sparsity, fact tables usually make up 90 percent or more of the total space consumed by a dimensional model. **Fact tables tend to be deep in terms of the number of rows, but narrow in terms of the number of columns.** Given their size, you should be judicious about fact table space utilization.

All fact tables have two or more foreign keys (refer to the FK notation in Figure 1-2) that connect to the dimension tables' primary keys. For example, the product key in the fact table always matches a specific product key in the product dimension table. When all the keys in the fact table correctly match their respective primary keys in the corresponding dimension tables, the tables satisfy referential integrity. You access the fact table via the dimension tables joined to it.

Retail Sales Facts
Date Key (FK)
Product Key (FK)
Store Key (FK)
Promotion Key (FK)
Customer Key (FK)
Clerk Key (FK)
Transaction #
Sales Dollars
Sales Units

Figure 1-2

The fact table generally has its own primary key composed of a subset of the foreign keys. This key is often called a composite key. Every table that has a composite key is a fact table. Fact tables express many-to-many relationships. All others are dimension tables.

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DIMENSIONAL MODELLING: **DIMENSION TABLES**

Dimension Tables are integral companions to a fact table. The dimension tables contain the textual context associated with a business process measurement event. They describe the "who, what, where, when, how, and why" associated with the event.

Dimension tables often have many columns or attributes. It is not uncommon for a dimension table to have 50 to 100 attributes; although, some dimension tables naturally have only a handful of attributes. Dimension tables tend to have fewer rows than fact tables, but can be wide with many large text columns. **Each dimension is defined by a single primary key (refer to the PK notation in Figure 1-3), which serves as the basis for referential integrity with any given fact table to which it is joined**.

Dimension attributes serve as the primary source of query constraints, groupings, and report labels. In a query or report request, attributes are identified as the by words. For example, when a user wants to see dollar sales by brand, brand must be available as a dimension attribute.

Product Dimension
Product Key (PK)
SKU Number (Natural Key)
Product Description
Brand Name
Category Name
Department Name
Package Type
Package Size
Abrasive Indicator
Weight
Weight Unit of Measure
Storage Type
Shelf Life Type
Shelf Width
Shelf Height
Shelf Depth
...

Figure 1-3

Figure on the right shows that **dimension tables often represent hierarchical relationships**. For example, products roll up into brands and then into categories. For each row in the product dimension, you should store the associated brand and category description. The hierarchical descriptive information is stored redundantly in the spirit of ease of use and query performance.

Product Key	Product Description	Brand Name	Category Name
1	PowerAll 20 oz	PowerClean	All Purpose Cleaner
2	PowerAll 32 oz	PowerClean	All Purpose Cleaner
3	PowerAll 48 oz	PowerClean	All Purpose Cleaner
4	PowerAll 64 oz	PowerClean	All Purpose Cleaner
5	ZipAll 20 oz	Zippy	All Purpose Cleaner
6	ZipAll 32 oz	Zippy	All Purpose Cleaner
7	ZipAll 48 oz	Zippy	All Purpose Cleaner
8	Shiny 20 oz	Clean Fast	Glass Cleaner
9	Shiny 32 oz	Clean Fast	Glass Cleaner
10	ZipGlass 20 oz	Zippy	Glass Cleaner
11	ZipGlass 32 oz	Zippy	Glass Cleaner

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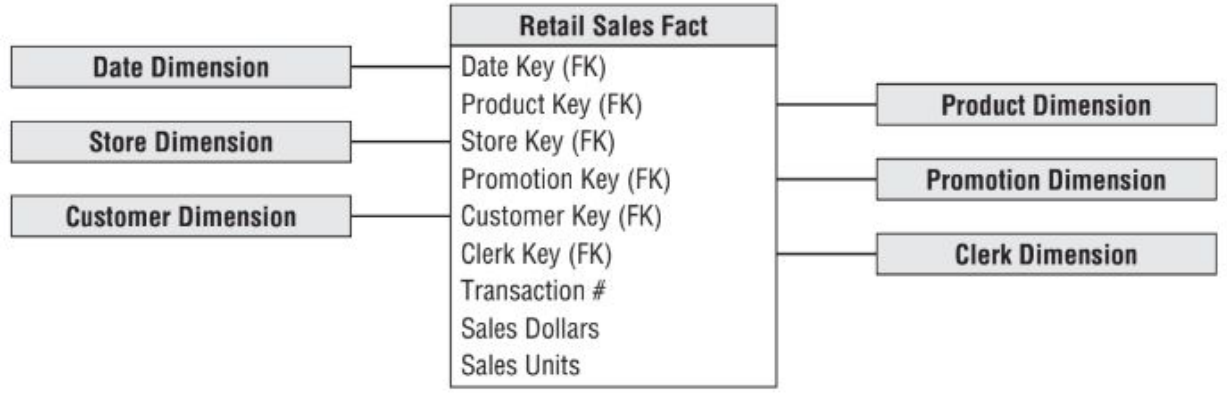
DIMENSIONAL MODELLING: **BRINGING BOTH**

Now that you understand fact and dimension tables, it's time to bring the building blocks together in a dimensional model, as shown in Figure 1-5. Each business process is represented by a dimensional model that consists of a fact table containing the event's numeric measurements surrounded by a halo of dimension tables that contain the textual context that was true at the moment the event occurred. This characteristic star-like structure is often called a **Star Join**, an alternatively known as **Star Schema**

The first thing to notice about the dimensional schema is its simplicity and symmetry. Obviously, business users benefit from the simplicity because the data is easier to understand and navigate.

The charm of the design in Figure 1-5 is that it is highly recognizable to business users. We have observed literally hundreds of instances in which users immediately agree that the dimensional model is their business. Furthermore, the reduced number of tables and use of meaningful business descriptors make it easy to navigate and less likely that mistakes will occur.



The simplicity of a dimensional model also has performance benefits. Database optimizers process these simple schemas with fewer joins more efficiently.

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DIMENSIONAL MODELLING: **BRINGING BOTH**

Another way to think about the complementary nature of fact and dimension tables is to see them translated into a report. As illustrated in Figure 1-6, dimension attributes supply the report filters and labelling, whereas the fact tables supply the report's numeric values.

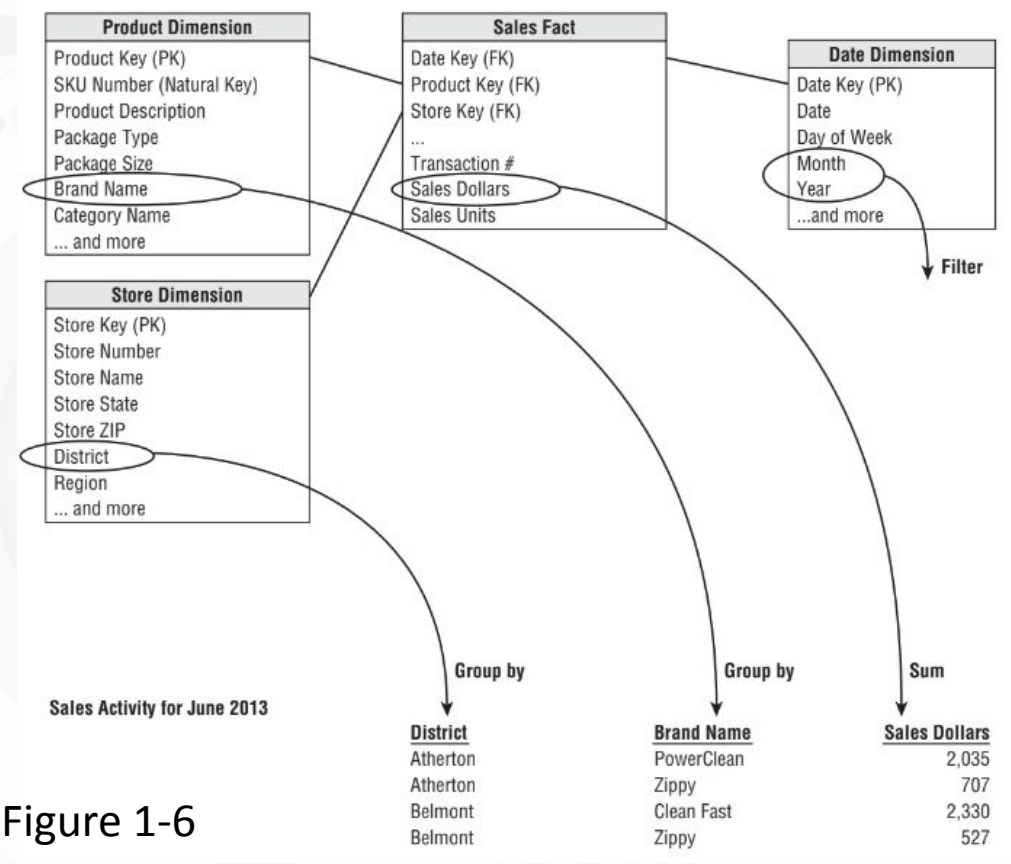

Figure 1-6

```
SELECT
        store.district_name,
        product.brand,
        sum(sales_facts.sales_dollars) AS "Sales Dollars"
FROM
        store,
        product,
        date,
        sales_facts

WHERE
                date.month_name="January" AND
                date.year=2013 AND

                -- JOINING CONDITIONS
                store.store_key = sales_facts.store_key AND
                product.product_key = sales_facts.product_key AND
                date.date_key = sales_facts.date_key

GROUP BY
                store.district_name,
                product.brand
```

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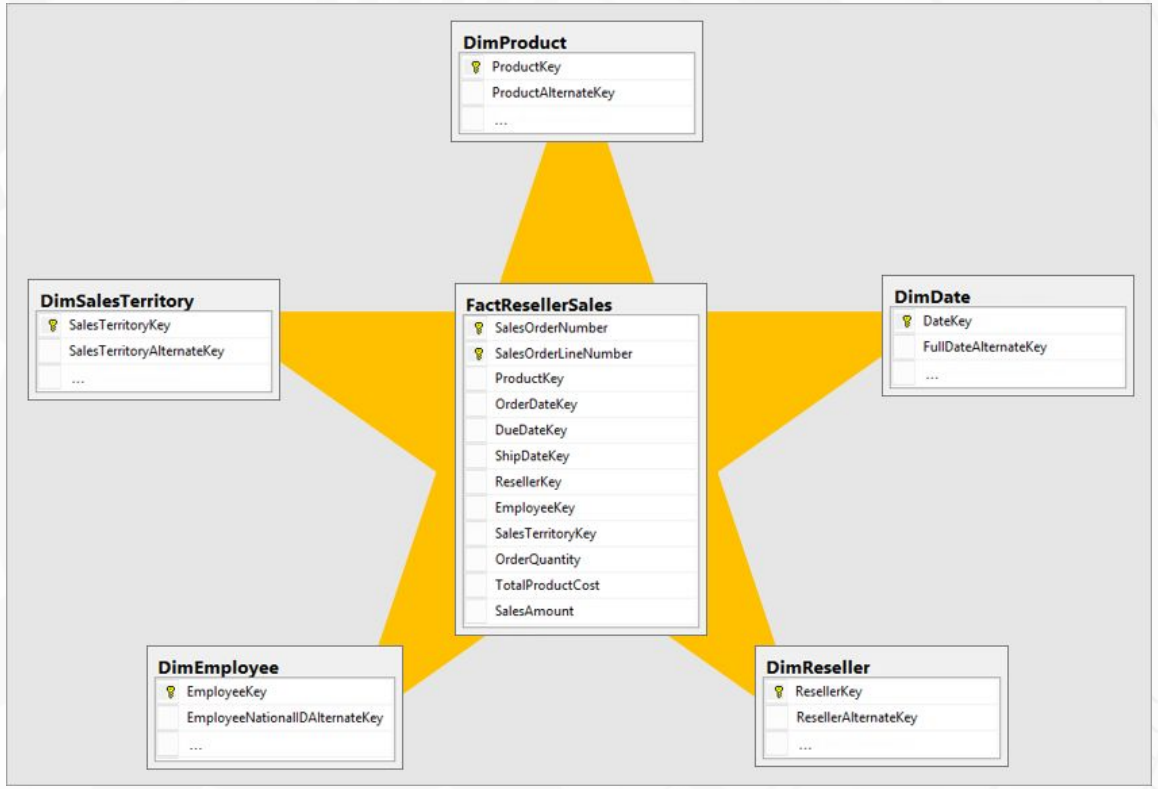
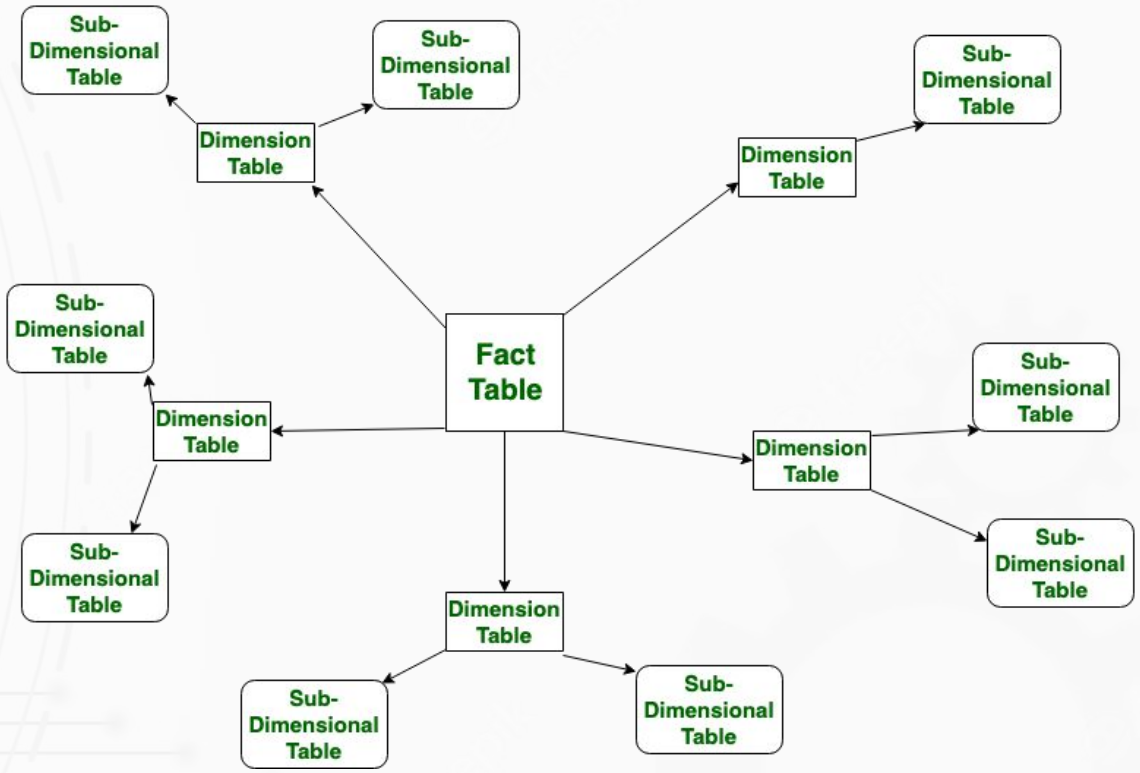
DIMENSIONAL MODELLING: **OTHER SCHEMAS**

STAR SCHEMA



SNOWFLAKE SCHEMA



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DIMENSIONAL MODELLING: **OTHER SCHEMAS**

STAR SCHEMA	SNOWFLAKE SCHEMA
Hierarchies for the dimensions are stored in the dimensional table.	Hierarchies are divided into separate tables.
It contains a fact table surrounded by dimension tables.	One fact table surrounded by dimension table which are in turn surrounded by dimension table
In a star schema, only single join creates the relationship between the fact table and any dimension tables.	A snowflake schema requires many joins to fetch the data.
Simple DB Design.	Very Complex DB Design.
Denormalized Data structure and query also run faster.	Normalized Data Structure.
High level of Data redundancy	Very low-level data redundancy
Single Dimension table contains aggregated data.	Data Split into different Dimension Tables.
Cube processing is faster.	Cube processing might be slow because of the complex join.
Offers higher performing queries using Star Join Query Optimization. Tables may be connected with multiple dimensions.	The Snow Flake Schema is represented by centralized fact table which unlikely connected with multiple dimensions.

https://www.guru99.com/star-snowflake-data-warehousing.html

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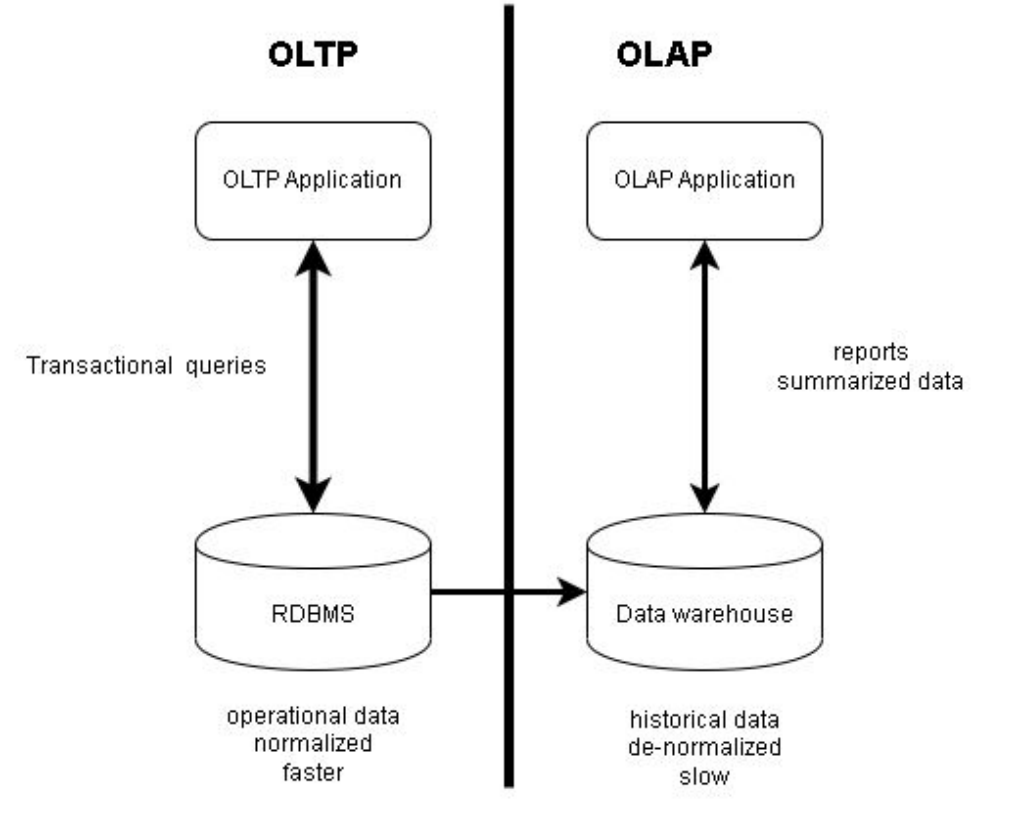
DIMENSIONAL MODELLING: **A GOOD DWH**

These are the operational systems of record that capture the business's transactions. Think of the source systems as outside the data warehouse because presumably you have little or no control over the content and format of the data in these operational systems. The main priorities of the source systems are processing performance and availability.

Operational queries against source systems are narrow, one-record-at-a-time queries that are part of the normal transaction flow and severely restricted in their demands on the operational system. It is safe to assume that source systems are not queried in the broad and unexpected ways that DW/BI systems typically are queried.

Source systems maintain little historical data; a good data warehouse can relieve the source systems of much of the responsibility for representing the past.

In many cases, the source systems are special purpose applications without any commitment to sharing common data such as product, customer, geography, or calendar with other operational systems in the organization.



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DM LAB: **DESIGNING STAR SCHEMA IN OLAP**

The Retail case

- The retail company has 100 grocery stores spread over a five state area
- Each store has a number of departments, including grocery, frozen food, dairy, meat, produce, bakery, floral, health/beauty aids
- Each store has roughly 60 000 individual products
- The individual products are called Stock Keeping Units (SKU)

Management is concerned with:

The logistics of ordering (the procurement are often done too late) the stock levels (too high stock levels for some products and too low for other) the selling of products (too low sales figure)

We can collect data:
at the back door of retail stores, where deliveries are made
at the depository where we keep our stock
at the cash registers, called point of sale (PoS), where customer purchases products.

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DM LAB: **DESIGNING STAR SCHEMA IN OLAP**

Our 4 Step Approach : KIMBALL

1. Select a Business Process
2. Declare the Grain
3. Identify the Dimensions
4. Identify the Facts

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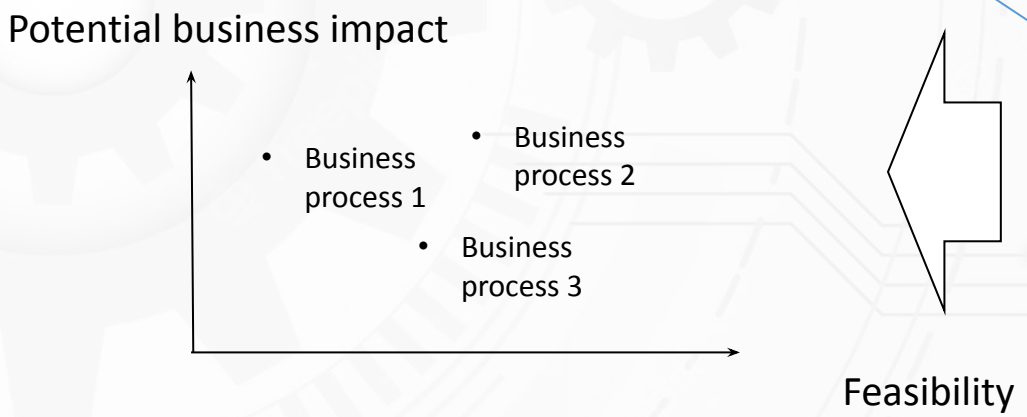
DM LAB: DESIGNING STAR SCHEMA IN OLAP

Step 1. Select a business process

- A business process is a “low-level activity” performed by an organization, such as taking orders, receiving payments, etc
- Kimball: ”The first dimensional model built should be the one with the most impact – it should answer the most pressing business questions and be readily accessible for data extraction.”



A dimensional model is based on a business process, therefore the focus on business process

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DM LAB: DESIGNING STAR SCHEMA IN OLAP

Step 1. Select a business process

- Which Business process should we select first?
    - the *logistics of ordering* can be interpreted as *the order process*
    - the *stock levels* can be interpreted as *the inventory process*
    - the *selling of products* can be interpreted as *the sales process*

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DM LAB: DESIGNING STAR SCHEMA IN OLAP

Step 1. Select a business process

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  - **the *selling of products* can be interpreted as *the sales process***

More information is needed to select the business process with most impact and highest feasibility. In this case, we assume that a sales process in an important process since the management is concerned with the low sales figures, and increased sales will probably make a larger impact than the other two processes

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DM LAB: DESIGNING STAR SCHEMA IN OLAP

Step 2. Declare the Grain

- The grains is the level of detail of data of the process or event. The grain means specifying exactly what an individual fact table row represent.
- Typical grains are individual transactions, individual daily (or monthly) snapshots

Kimball:

”The less granular model is immediately vulnerable to unexpected user requests to drill down into the detail.”

“Preferably you should develop dimensional models for the most atomic information captured by a business process.”

This is how you can express the most detailed level of data:

- Most granular data (the more you go into details, the more granular data you have)
- High granularity (the more you go into details, the higher granularity you have)

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DM LAB: DESIGNING STAR SCHEMA IN OLAP

Step 2. Declare the Grain

- What grain shall we choose for our case study?
    - An individual line item on a POS transaction, i.e. the sale of a product, per store, per transaction)
    - The sale of a product, per store, per day
    - The sale of a product, per region, per day
    - The sale of a product, per store, per month
    - The sale of a product category, per region, per month

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DM LAB: DESIGNING STAR SCHEMA IN OLAP

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  - The sale of a product, per region, per day
  - The sale of a product, per store, per month
  - The sale of a product category, per region, per month

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DM LAB: DESIGNING STAR SCHEMA IN OLAP

Step 3. Identify the Dimensions

- Identify the dimensions that will be applied to each fact table tuple/record
  - Consider what aspects do you want to be able to slice and dice
  - Typical dimensions are date, product, customer, store, etc
- Guideline: Use the declared grain of the selected process to determine dimensions: product, store
- Guideline: Use the Who, What, When, Where, and How to identify possible dimensions

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Dimensions : **Product**

- Use surrogate key as primary key - to keep history in if natural keys are changing in the organization
- How to identify attributes:
  - usually text fields, with discrete values, e.g., the flavor of a product, the size of a product
- Denormalization is no problem since the ETL process ensure the data quality.
- Use column names/labels that are easy to understand
- Use names/labels of the values that are easy to understand

The end users will see these names/labels interacting with the system using an OLAP tool

Product

ProductKey	SKU	Product Name	Product Description	Supplier	Product Class	Product Property	Package Type	Volume	Height	Widht	Depth	Height-Widtht-Depth	Fat content
1	7 310865 001191	Blue Milk	Milk with vitamin D and 0,5% fat	Arla	Dairy	Low fat	Roof ridge pack	1 liter	23 cm	7 cm	7 cm	23x7x7	0,5 %
2	7 310865 001795	Green Milk	Milk with vitamin D and 1,5% fat	Arla	Dairy	Plain	Brick pack	1 liter	16 cm	6,5 cm	9,5 cm	16x6,5x 9,5	1,5 %
3	7 310865 000156	Red Milk	Milk with vitamin D and 3% fat	Arla	Dairy	Plain	Brick pack	1 liter	16 cm	6,5 cm	9,5 cm	16x6,5x 9,5	3 %
4	7 300170 062642	Green Milk	Local Milk with vitamin D and 1,5% fat	Coop Änglam ark	Dairy	Ecological	Roof ridge pack	1 liter	23 cm	7 cm	7 cm	23x7x7	1,5 %

Product_D

ProductKey (PK)
Product_Name
Product Description
Product property
Product class
SKU_Number (NK)
Supplier
Package_Type
Volume
Height
Width
Depth
Height-Weight-Depth
Fat_Content

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Dimensions : **Store**

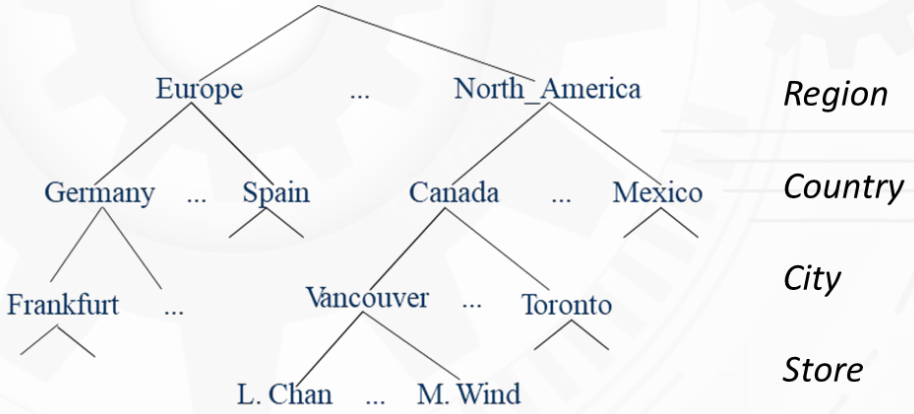
Another dimension: Store dimension

- A geographic dimension - every store has a location
- Dimensions have hierarchies - we can roll up and drill down on geographic attributes

Concept: Dimensional hierarchies

A hierarchy is a directed tree which nodes are dimensional attribute values/elements

There is a many-to-one relationship between the nodes in the hierarchy



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Dimensions : **Date**

- **Date dimension – most star schema has a date dimension**
- The data can be populated in advance.
- It has one meaningful key (intelligent/smart key). That is, it does not have any surrogate key due to performance, ease of use, partitioning reasons
- A **fiscal year** (or **financial year**, or sometimes **budget year**) is the period used by governments for accounting and budget purposes, which varies between countries. It is also used for financial reporting by business and other organizations.
- Laws in many jurisdiction require company financial reports to be prepared and published on an annual basis, but generally do *not* require the reporting period to align with the calendar year (1 January to 31 December). (Wikipedia)

Date Key	Date	Full Date Description	Day of Week	Calendar Month	Calendar Quarter	Calendar Year	Fiscal Year Month	Holiday Indicator	Weekday Indicator
20130101	01/01/2013	January 01, 2013	Tuesday	January	Q1	2013	F2013-01	Holiday	Weekday
20130102	01/02/2013	January 02, 2013	Wednesday	January	Q1	2013	F2013-01	Non-Holiday	Weekday
20130103	01/03/2013	January 03, 2013	Thursday	January	Q1	2013	F2013-01	Non-Holiday	Weekday
20130104	01/04/2013	January 04, 2013	Friday	January	Q1	2013	F2013-01	Non-Holiday	Weekday
20130105	01/05/2013	January 05, 2013	Saturday	January	Q1	2013	F2013-01	Non-Holiday	Weekday
20130106	01/06/2013	January 06, 2013	Sunday	January	Q1	2013	F2013-01	Non-Holiday	Weekday
20130107	01/07/2013	January 07, 2013	Monday	January	Q1	2013	F2013-01	Non-Holiday	Weekday
20130108	01/08/2013	January 08, 2013	Tuesday	January	Q1	2013	F2013-01	Non-Holiday	Weekday

Date Dimension

Date Key (PK)
Date
Full Date Description
Day of Week
Day Number in Calendar Month
Day Number in Calendar Year
Day Number in Fiscal Month
Day Number in Fiscal Year
Last Day in Month Indicator
Calendar Week Ending Date
Calendar Week Number in Year
Calendar Month Name
Calendar Month Number in Year
Calendar Year-Month (YYYY-MM)
Calendar Quarter
Calendar Year-Quarter
Calendar Year
Fiscal Week
Fiscal Week Number in Year
Fiscal Month
Fiscal Month Number in Year
Fiscal Year-Month
Fiscal Quarter
Fiscal Year-Quarter
Fiscal Half Year
Fiscal Year
Holiday Indicator
Weekday Indicator
SQL Date Stamp
...

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DM LAB: DESIGNING STAR SCHEMA IN OLAP

Step 4. Identify the Facts

- Establishing what we want to measure, that is, the process metrics or key performance indicators (KPI)
- How to identify the facts?
  - “something not known in advance” (Note, the values of the dimensions we know in advance)
  - an observation
  - many facts (but not all) have numerical values

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DM LAB: DESIGNING STAR SCHEMA IN OLAP

Step 4. Identify the Facts

- What facts shall we choose for our case study?
- What can we find out of a PoS?
    - The number of products?
    - The total width of the products?
    - The total sum of the products

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DM LAB: DESIGNING STAR SCHEMA IN OLAP

Step 4. Identify the Facts

- What facts shall we choose for our case study?
- What can we find out of a PoS?
  - **The number of products?** → Sales Quantity
  - The total width of the products?
  - **The total sum of the products** → Sales Dollar Amount

Point of Sales F			
Product Key	:	Integer	(PK, FK)
DateKey	:	Integer	(PK, FK)
StoreKey	:	Integer	(PK, FK)
Sales Quantity	:	Integer	(fact)
Sales Dollar Amount	:	Integer	(fact)

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DM LAB: DESIGNING STAR SCHEMA IN OLAP

Step 4. Identify the Facts

There are two kinds of fact table attributes:

- **Classification attributes:** Keys that captures the dimensions of the process
  - Keys are usually represented in integer form and do not require much memory space

- **Facts:** Attributes that captures the measures of the process
  - Facts are often numeric properties and can usually be represented as integers
  - Contrast to dimensional attributes which are usually long text strings

The Primary key for the fact table is a concatenated key consisting of the foreign keys to (several of) its dimensions

Point of Sales F			
Product Key	:	Integer	(PK, FK)
DateKey	:	Integer	(PK, FK)
StoreKey	:	Integer	(PK, FK)
Sales Quantity	:	Integer	(fact)
Sales Dollar Amount	:	Integer	(fact)

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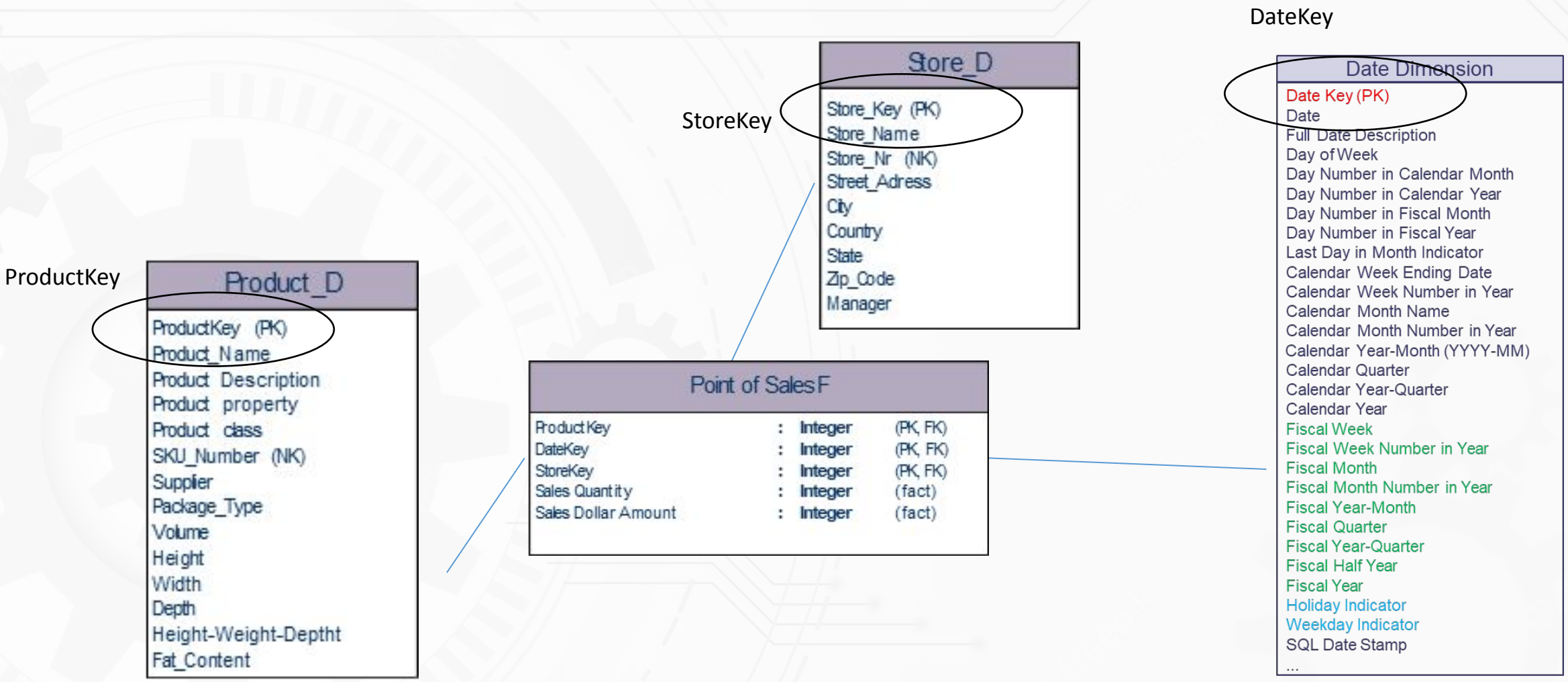
DM LAB: DESIGNING STAR SCHEMA IN OLAP

THE FINAL STAR SCHEMA



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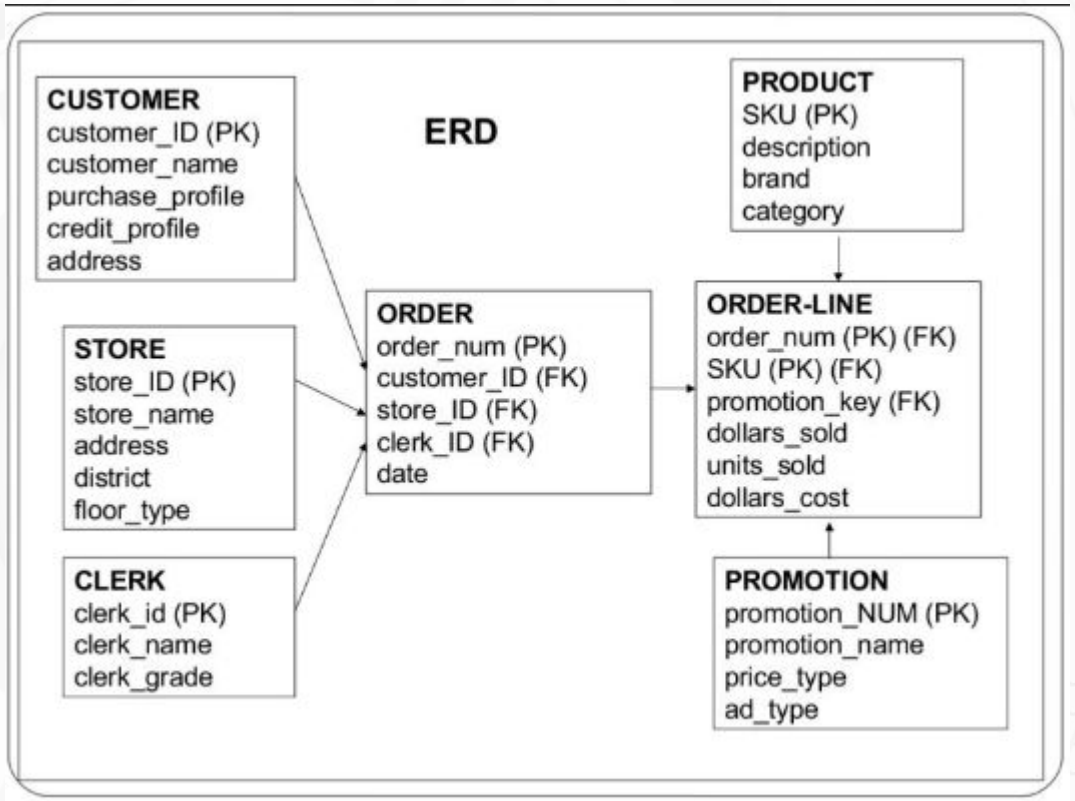
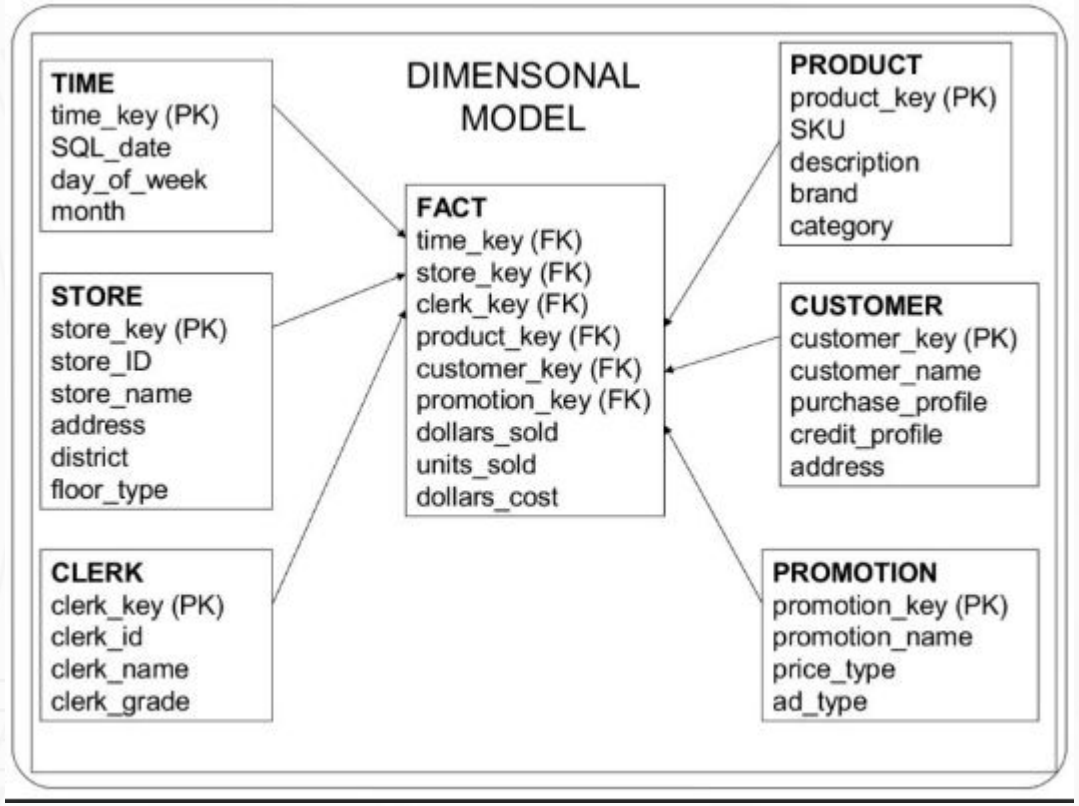
DM LAB: DESIGNING STAR SCHEMA IN OLAP

ANOTHER QUICK EXAMPLE

OLTP NORMALIZED DATABASE



OLAP STAR SCHEMA



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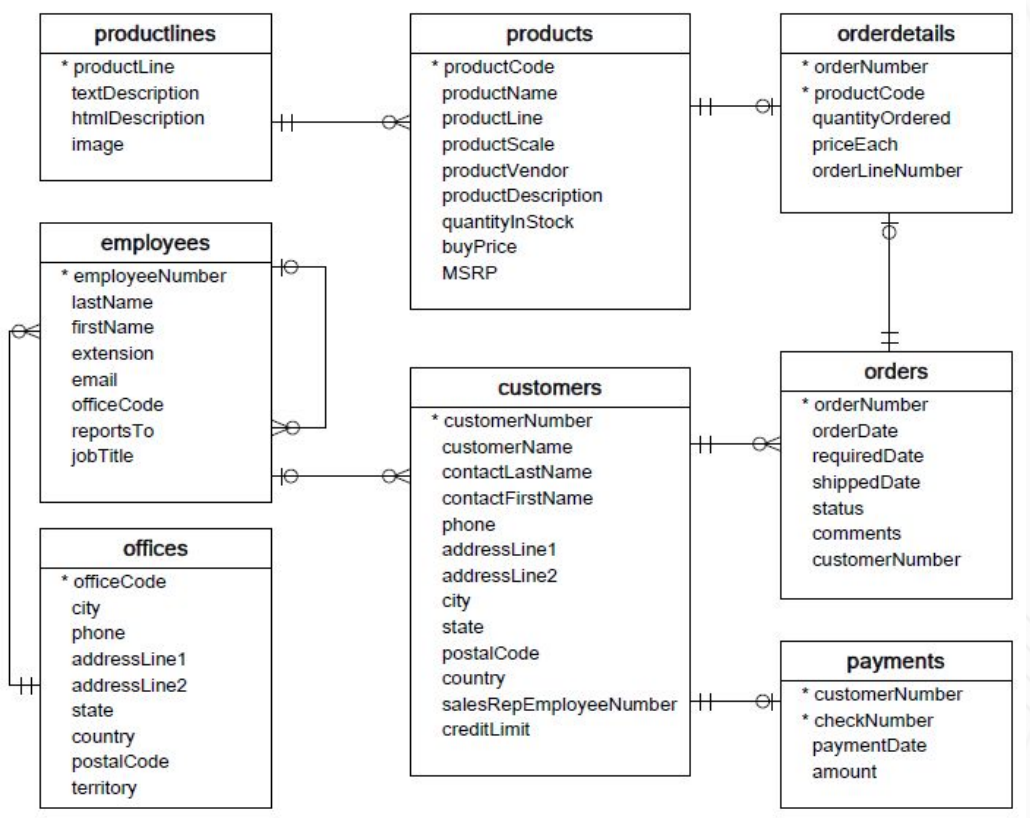
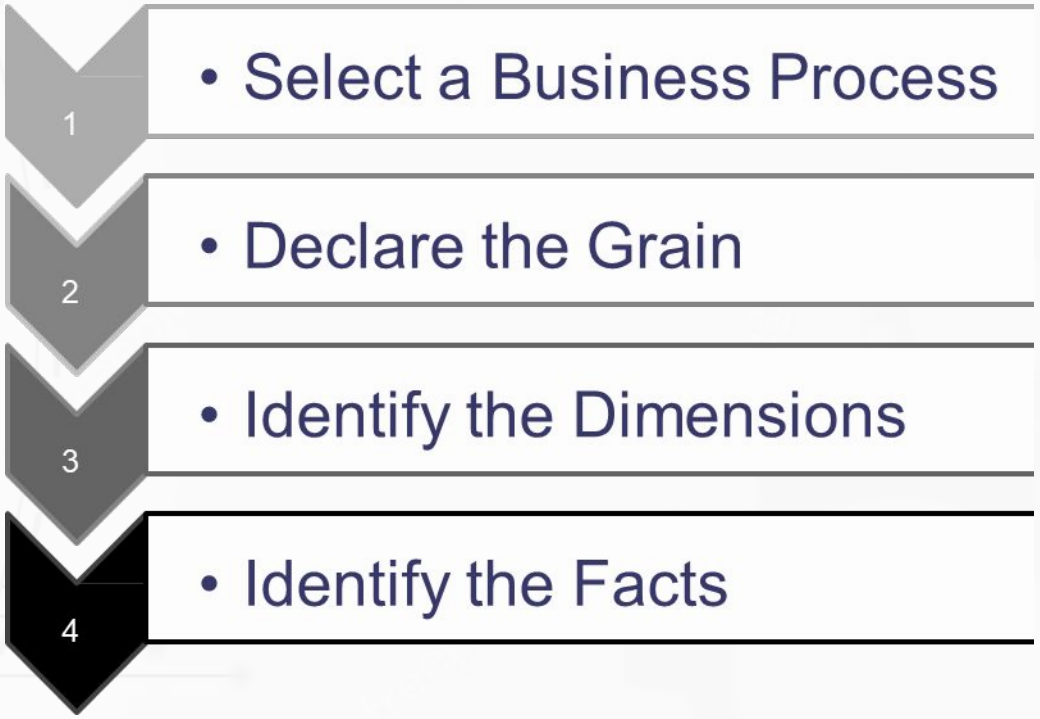
DM LAB: DESIGNING STAR SCHEMA IN OLAP

CLASS ACTIVITY SQL, CLASSIC MODELS OLAP CREATION

OLTP NORMALIZED DATABASE



OLAP STAR SCHEMA



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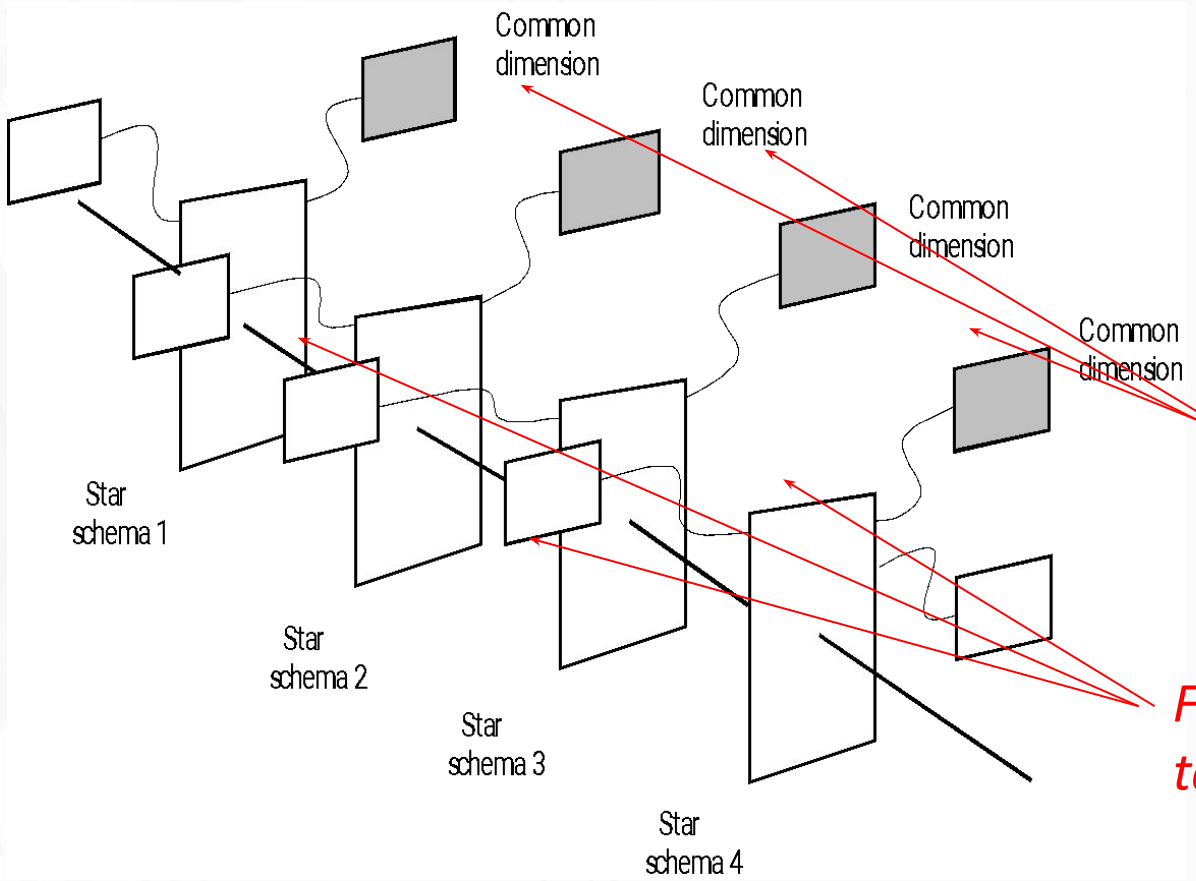
DIMENSIONAL MODELLING: **THE UNIVERSE**

A family of stars



Dimension tables (in this case in the figure the same dimension for all fact tables)

Fact tables (different fact tables)

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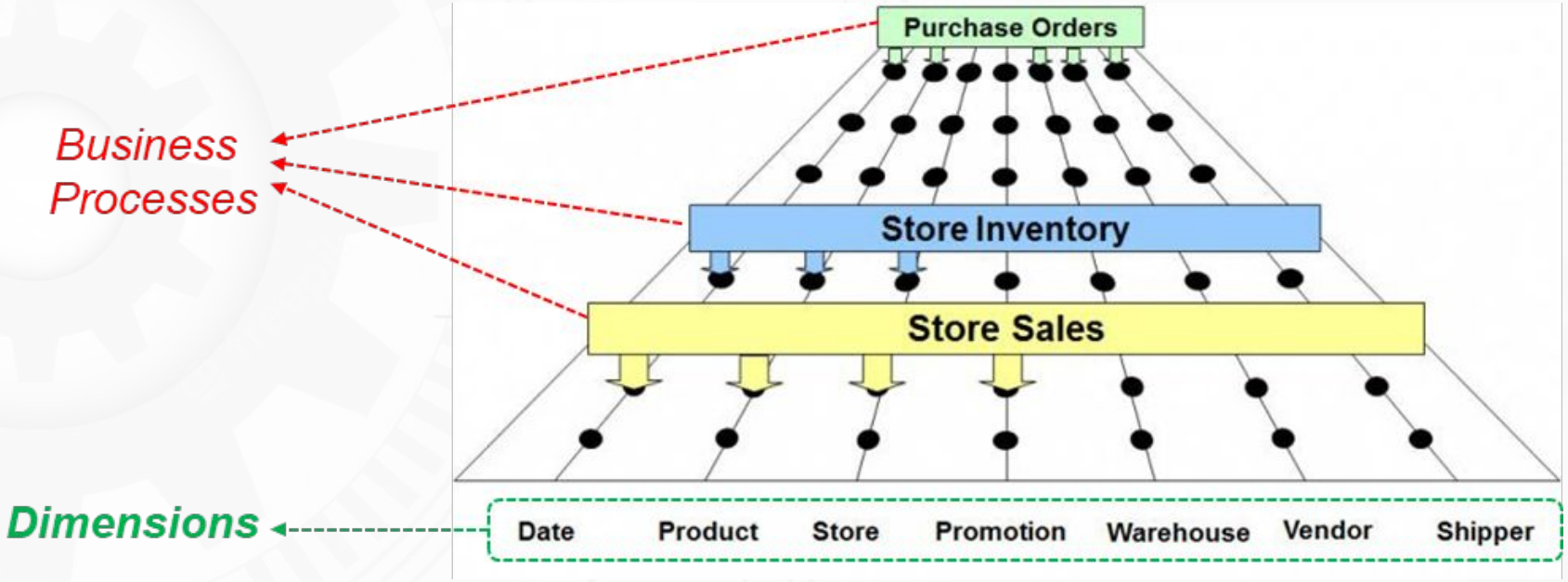
DIMENSIONAL MODELLING: **THE UNIVERSE**

There are many processes

- There are many business processes. Therefore, we need many stars schemas. Therefore, We need an enterprise architecture, called the **Enterprise Data Warehouse Bus Architecture**



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DIMENSIONAL MODELLING: **THE UNIVERSE**

Enterprise Data Warehouse Bus Matrix

- Enterprise Data Warehouse Bus Matrix
  - Rows represent business processes
  - Columns represent dimensions
  - A dimension that is shared among several business processes is called conformed dimension
  - It is a good idea to create a list of core dimensions early

- Benefits:
  - The matrix delivers the big picture perspective, regardless of database or technology preferences.

	COMMON DIMENSIONS						
BUSINESS PROCESSES	Date	Product	Warehouse	Store	Promotion	Customer	Employee
Issue Purchase Orders	X	X	X				
Receive Warehouse Deliveries	X	X	X				X
Warehouse Inventory	X	X	X				
Receive Store Deliveries	X	X	X	X			X
Store Inventory	X	X		X			
Retail Sales	X	X		X	X	X	X
Retail Sales Forecast	X	X		X			
Retail Promotion Tracking	X	X		X	X		
Customer Returns	X	X		X	X	X	X
Returns to Vendor	X	X		X			X
Frequent Shopper Sign-Ups	X			X		X	X

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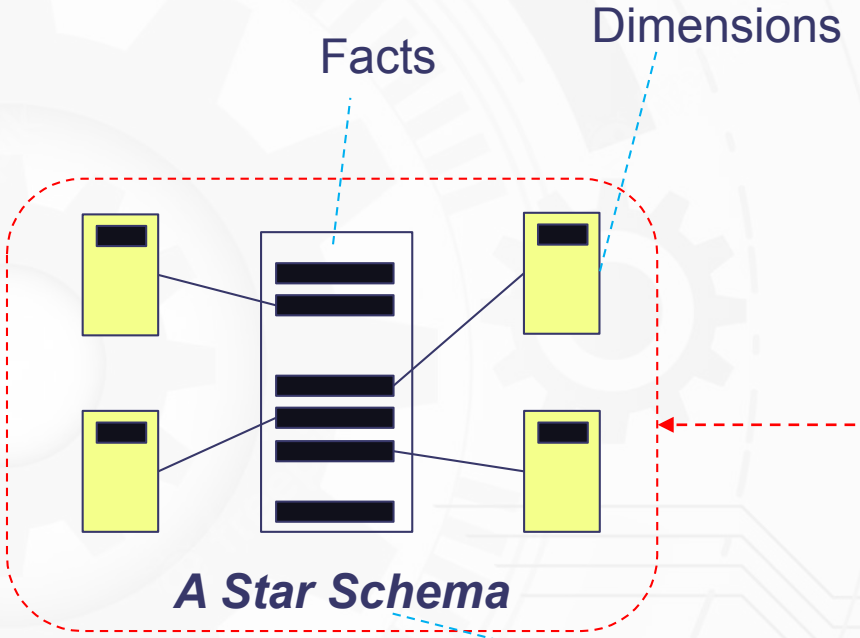
DIMENSIONAL MODELLING: **THE UNIVERSE**

Enterprise Data Warehouse Bus Matrix



BUSINESS PROCESSES	COMMON DIMENSIONS						
	Date	Product	Warehouse	Store	Promotion	Customer	Employee
Issue Purchase Orders	X	X	X				
Receive Warehouse Deliveries	X	X	X				X
Warehouse Inventory	X	X	X				
Receive Store Deliveries	X	X	X	X			X
Store Inventory	X	X		X			
Retail Sales	X	X		X	X	X	X
Retail Sales Forecast	X	X		X			
Retail Promotion Tracking	X	X		X	X		
Customer Returns	X	X		X	X	X	X
Returns to Vendor	X	X		X			X
Frequent Shopper Sign-Ups	X			X		X	X

The star schema is designed based on one business process

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DIMENSIONAL MODELLING: **THE UNIVERSE**

Enterprise Data Warehouse Bus Matrix

- Conformed dimensions – the same dimensions are used by several facts
- Thereby, it possible to join different star schemas if needed

	COMMON DIMENSIONS						
BUSINESS PROCESSES	Date	Product	Warehouse	Store	Promotion	Customer	Employee
Issue Purchase Orders	X	X	X				
Receive Warehouse Deliveries	X	X	X				X
Warehouse Inventory	X	X	X				
Receive Store Deliveries	X	X	X	X			X
Store Inventory	X	X		X			
Retail Sales	X	X		X	X	X	X
Retail Sales Forecast	X	X		X			
Retail Promotion Tracking	X	X		X	X		
Customer Returns	X	X		X	X	X	X
Returns to Vendor	X	X		X			X
Frequent Shopper Sign-Ups	X			X		X	X

Therefore, we can say that Date Product & Store are confirmed dimensions, since they are referred by multiple fact representing multiple business processes.

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ADVANCE DIMENSIONAL MODELLING

TYPES OF FACTS

1. Transaction Fact Table

This is a fundamental and basic view of business operations. It is used to represent an occurrence of an event at any instantaneous point in time. The facts measure are valid only for that particular instant and only for that event. The grain which is associated with the transaction table specifies as “one row per line in a transaction”. Usually, it contains the data of the detailed level, which leads it to have a large number of dimensions associated with it. It captures the measurement at the most basic or atomic level of dimension. This helps the table to give robust dimensional grouping, roll up & drill-down reporting capabilities to the users. It is dense and sparse. It can be large, maybe containing billions of records. Let us see an example of sales in a grocery shop.

Transactional Facts

SHOP_ID	EMP_ID	CUSTOMER_ID	PRODUCT_ID	SALES	QUANTITY	CURRENCY
100	50	900	10	16	2	USD
100	60	900	1209	100	1	USD
101	50	1009	3459	50	9	USD
102	70	1007	56	250	4	USD

Dimensions

Measures, KPIs Units

One record represent one transaction, or one event at a time

2. Snapshot Fact Table

The snapshot gives the state of things at a particular instance of time or “picture of the moment”. It normally includes more non-additive and semi-additive facts. It helps to review the cumulative performance of the business at regular and predictable intervals of time. In this, the performance of an activity at the end of each day or a week or a month or any other time interval is represented, unlike the transaction fact table where a new row is added for the occurrence of every event. But snapshot fact tables or periodic snapshots are dependent on the transaction fact table to get the detailed data present in the transaction fact table. The periodic snapshot tables are mostly dense and can be large as transaction fact tables. Let us see an example of the periodic snapshot of the sales of the same grocery shop as in the transaction fact table.

CLINIC_ID	WEEK	VISITS	SURGERIES	CONSULTATION
100	W1	30	2	20
100	W2	60	3	8
101	W1	25	4	6
102	W1	40	1	4
101	W3	0	0	0

3. Accumulating Fact Tables

These are used to represent the activity of any process that has a well defined and clear starting and end. Accumulating snapshots mostly have multiple data stamps that represent the predictable phases or events that occur during the lifetime. Sometimes there is an extra column containing the date that shows when the row was last updated. Let us see an example.

COMPLAIN_ID	ASSIGNED_EMP	START_DT	STATUS	INVEST_DT	RESOLUTION_DT	REVIEW_DT
100	100	02082019	3	03082019	04082019	05082019
101	230	02102019	0			
102	435	15082018	0			
103	596	02112019	0			
104	294	11072019	0			

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ADVANCE DIMENSIONAL MODELLING

TYPES OF FACTS

1. Transaction Fact Table

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ADVANCE DIMENSIONAL MODELLING

CHANGE DIMENSION HISTORY

SCD 0: No Change

What's the use?

A Date Dimension is an SCD 0 It will never change. It's a Universal Fact :-D

Or Some Attributes in an Existing Dimension may be SCD 0 Like Never Changing

Employee Hire
Date and **DateFirstPurchase** are the attributes that will not be updated which is Type 0 SCD.

OLTP

ID	NAME	DOB	DEPARTMENT	.....
101	Aman	13-09-1993	HR	
102	Rajeev	02-03-1995	SAILS	
103	Prabhat	23-04-1990	IT	

OLAP

ID	NAME	DOB	DEPARTMENT	.....
101	Aman	13-09-1993	HR	
102	Rajeev	03-02-1995	SAILS	
103	Prabhat	23-04-1990	IT	

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ADVANCE DIMENSIONAL MODELLING

CHANGE DIMENSION HISTORY

SCD 1: Update

The Type 1 method is used when no historical data is required on how data changed in the SCD field over time. Type 1 method simply updates a record in the dimension table, overwriting an old SCD value with a new value.

When to use SCD Type 1?

- Usually, the changes relate to correction of errors in the source system
- Sometimes the change in the source system has no significance
- The old value in the source system needs to be discarded
- The change in the source system need not be preserved in the DWH

Before Update

CustomerKey	Customer Surrogate Key	CompanyName	ContactName	ContactTitle
ANATR	1	The Digital	Aaron Higgs	Sales Representative

After Update

CustomerKey	Customer Surrogate Key	CompanyName	ContactName	ContactTitle
ANATR	1	The Digital	Aaron Higgs	Business Development Manager

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ADVANCE DIMENSIONAL MODELLING

CHANGE DIMENSION HISTORY

SCD 1: Update

How to Implement SCD Type 1?

In our fictitious scenario, the business is only interested in preserving history for ContactName for their primary contact at each customer business, while associated details for any ContactName require an overwrite because the business is only interested in tracking historical data for their primary contact. SCD1 – Update option is therefore selected for ContactTitle and Fax columns.

We have updated only the ContactTitle here and are not interested in preserving history here. As this column has SCD field type set to 'SCD1 – Update', Procedure has overwritten the previous value with the new one.

Before Update

CustomerKey	Customer Surrogate Key	CompanyName	ContactName	ContactTitle
ANATR	1	The Digital	Aaron Higgs	Sales Representative

After Update

CustomerKey	Customer Surrogate Key	CompanyName	ContactName	ContactTitle
ANATR	1	The Digital	Aaron Higgs	Business Development Manager

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ADVANCE DIMENSIONAL MODELLING

CHANGE DIMENSION HISTORY

SCD 1: Update

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